Microelectronics

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October 17, 2026

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1 Microelectronics

I Signal

i Signal

A signal is a function that conveys information about a phenomenon. Signal processing is usually most conveniently performed by electronic systems on voltage or current signals, especially the former. Only voltage and current signals are studied here.

ii Signal source

A signal source is typically represented in the Thévenin form, a voltage source in a series connection with a resistance, or the Norton form, a current source in a parallel connection with a resistance.

iii Cascade

Output of a component as the input of another.

iv Transducer

A transducer is a device that convert a signal to a voltage or current signal or vice versa.

v Signal processing

Signal processing includes analog signal processing (ASP), analog-to-digital converter (ADC), digital signal processor (DSP), digital-to-analog converter (DAC), amplifier, etc.

vi Coupling capacitor

A capacitor connected in series between stages to pass AC signal while blocking DC.

vii Symbol convention

- Lowercase character with uppercase subscript means total signal, e.g. $i_C(t)$.
- Lowercase character with lowercase subscript means AC signal, e.g. $i_c(t)$.
- Uppercase character with uppercase subscript means DC signal, e.g. I_C.
- Uppercase character with lowercase subscript means peak signal, e.g. I_c.

Their relation is

$$i_C(t) = i_c(t) + I_C$$

$$i_c(t) = I_c \sin(\omega t + \varphi)$$

viii Differential and common-mode signal

For two input signals, the differential signal (AC signal) is the difference between them, and the common-mode signal (DC signal) is the algebraic average of them.

ix Input and output impedance

For a component's input, input impedance is the impedance relative to ground as seemed from the input.

For a component's output, output impedance is the impedance relative to ground as seemed from the input.

x Differential and common-mode impedance

For two input terminals, the differential input impedance is the input impedance relative to ground as seemed by them when a ideal source connects them directly.

For two input terminals, the common-mode input impedance is the input impedance relative to ground as seemed by them when an ideal source's output connects both of them.

xi Differential and common-mode transfer function

For two input terminals, the differential transfer function is the transfer function when they are connected to $\frac{1}{2}$ differential input signal and $-\frac{1}{2}$ differential input signal respectively.

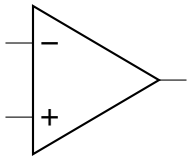
For two input terminals, the common-mode transfer function is the transfer function when an ideal source's output connecting both of them is treated as input signal.

II Amplifier (amp)

i Amplifier

An electronic device which amplifies an electrical signal by a scale called amplifier gain A and optionally convert voltage signal to current signal or vice versa.

ii Diagram



iii Two-port model

An amplifier can be modeled as a two port model, where an one input port and one output port is related by the amplifier, which acts as a controlled source controlled by the input port voltage or current which enforces voltage or current on the output port.

iv Circuit model of voltage amplifier

A voltage amplifier draws current from a source v_s with internal resistance R_s with input resistance R_i and voltage v_{in} across it and supplies $v_c = \mu v_{in}$ to a load network with output resistance R_o and load R_L, where μ, also denoted A_{vo}, is open-loop voltage gain factor.

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{v_c}{v_{in}} = \mu$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = \mu \frac{R_i R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\mu|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \ll R_L, \quad A_v \approx \mu$$

v Circuit model of current amplifier

A current amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $i_c = \beta i_{in}$ by CCCS to a load network with output resistance R_o in parallel and load R_L, where β, also denoted A_{cs}, is short-circuit current gain factor.

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{i_c}{i_{in}} = \beta$$

$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Current gain:

$$A_c = \frac{i_{out}}{i_{in}} = \beta \frac{R_s R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\beta|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \gg R_L, \quad A_c \approx \beta$$

vi Circuit model of transconductance amplifier

A transconductance amplifier draws current from a voltage source v_s with internal resistance R_s in series with input resistance R_i and current i_{in} through it and supplies $i_c = g v_{in}$ by VCCS to a load network with output resistance R_o in parallel and load R_L, where μ is open-loop transconductance factor.

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{i_c}{v_{in}} = g$$

$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Gain:

$$A_g = \frac{i_{out}}{v_{in}} = g\frac{R_i R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |g|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \gg R_L, \quad A_g \approx g$$

vii Circuit model of transresistance amplifier

A transresistance amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $v_c = ri_{in}$ by CCVS to a load network with output resistance R_o in series and load R_L, where r is short-circuit transresistance factor.

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{v_c}{i_{in}} = r$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Gain:

$$A_r = \frac{v_{out}}{i_{in}} = \beta\frac{R_s R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |r|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \ll R_L, \quad A_c \approx r$$

viii Amplifier gain in decibels (dB)

A power gain A_p in dB is defined as

$$G_p = 10\log_{10}(A_p)$$

A voltage gain A_v in dB is defined as

$$G_v = 20\log_{10}(A_v)$$

A current gain A_c in dB is defined as

$$G_c = 20\log_{10}(A_c)$$

ix Saturation

The linear characteristic only remains within a range. A positive and negative saturation levels can be observed.

x S parameter

If the energy enters through port i and emerges from port j, the S parameter is named S_{ji}.

xi Unilateral amplifier

An unilateral amplifier is an amplifier in which the signal transfer is effectively one-way, that is, the output signal is only dependent on the input signal but not what is connected to the output terminal. That is for each output port j and each input port i, $S_{ij} = 0$.

xii Power supply

To amplify signal, power supplies are needed, but often omitted in the diagram.

- V_{CC+} or V_{CC}: positive supply voltage, denoted on the + side of an op-amp or omitted;
- V_{CC-} or $-V_{EE}$: negative supply voltage, denoted on the − side of an op-amp or omitted.

The DC power delivered is the sum of power from the positive and negative power supply terminals V_{CC} and V_{EE}

$$P_{DC} = V_{CC}I_{CC} + V_{EE}I_{EE}$$

Suppose input and output signals have power P_I and P_L.

$$P_{DC} + P_I = P_L + P_{\text{dissipated}}$$

where $P_{\text{dissipated}}$ is the power dissipated.

xiii Classification based on frequency response

A band-pass amplifier is also called tuned amplifier. A low-pass amplifier is also called DC amplifier. A amplifier directly coupled with a coupling capacitor, called capacitively coupled, is called AC amplifier.

xiv Output impedance

In Laplace domain, for an amplifier with gain $A = \frac{V_o}{V_i}$ and output impedance Z_o driving a load impedance Z_L, the output voltage appears across the load V_L is:

$$V_L = V_o \frac{Z_L}{Z_o + Z_L},$$

and the gain as seemed from the load $A_L = \frac{V_L}{V_i}$ is:

$$A_L = A \frac{Z_L}{Z_o + Z_L},$$

and the power output to the load is

$$P_L = \frac{{V_L}^2}{Z_L}$$

xv Input impedance

In Laplace domain, for an amplifier with input impedance Z_i, the power input is:

$$P_i = \frac{{V_i}^2}{Z_i}$$

xvi Power gain

For an amplifier with gain A, input impedance Z_i, and output impedance Z_o driving a load impedance Z_L, the power gain (Laplace domain) is:

$$G_P = \frac{P_L}{P_i} = {A_L}^2 \frac{Z_i}{Z_L} = \frac{Z_i Z_L}{\left(Z_o + Z_L\right)^2}$$

III Operational Amplifier (op amp or op-amp)

i Definition

An op amp has five terminals:

- v_+: non-inverting input voltage of the terminal at the + sign;
- v_-: inverting input voltage of the terminal at the − sign;
- v_o: output voltage of the terminal at the opposite side of inputs;
- V_{CC+} or V_{CC}: positive supply voltage, denoted on the + side of an op-amp or omitted;
- V_{CC-} or $-V_{EE}$: negative supply voltage, denoted on the − side of an op-amp or omitted.

$$v_o = A(v_+ - v_- - v_{os})$$

where A is (differential) open-loop gain or open-circuit gain, and v_{os} is input offset voltage.

$$v_{ld} = v_+ - v_-$$
$$v_{lcm} = \frac{v_+ + v_-}{2}$$

ii Ideal op amp

- Infinite input impedance: No signal current flowing into v_+ and v_-.
- Infinite open-loop gain.
- Infinite bandwidth: The open-loop gain remains constant in all frequencies.
- Zero output impedance: The amplifier is unilateral.
- Zero input offset voltage: $v_{os} = 0$, which implies zero common-mode gain, i.e., infinite common-mode rejection.

This is a linear element with inputs v_+ and v_- and output v_o.

v_o may be any voltage since $A = \infty$ and $0 * \infty$ can be any number in $[0, \infty]$ in limit sense.

The open-loop gain of a real op amp is finite and dependent on frequency.

iii Internally compensated op amp

An internally compensated op amp is am op amp whose open-loop transfer function $A(s)$ is STC low-pass filter to ensure the op amp is stable,

$$A(s) = \frac{A_0}{1 + \frac{s}{\omega_b}} = \frac{\omega_t}{s + \omega_b}, \quad \omega_t = A_0 \omega_b,$$

where $A_0 \gg 1$ is DC gain, ω_b is 3 dB bandwidth, and ω_t is approximately the frequency of which the gain $|A|$ reach unity (0 dB):

$$|A(j\omega)| = \frac{A_0}{\sqrt{1 + \left(\frac{\omega}{\omega_b}\right)^2}} = 1$$

$$A_0{}^2 = 1 + \left(\frac{\omega}{\omega_b}\right)^2$$

$$\omega = \sqrt{A_0{}^2 - 1}\omega_b \approx A_0 \omega_b$$

The non-angular frequency $f_t = \frac{\omega_t}{2\pi}$ is usually specified on the op-amp data sheet as the unity-gain bandwidth. Such op amp is said to have a single-pole model since the single pole dominates the amplifier responses, also called dominant pole.

iv Feedback configuration

To have stable signal, op amp in a circuit is connected with a negative feedback configuration, which consists of the op amp and a connection from v_o to either of the input terminals with a resistance on it, called feedback connection. When the feedback connection connects to v_-, it is called inverting configuration. When the feedback connection connects to v_+, it is called non-inverting configuration.

We can construct a virtual short circuit between v_+ and v_- since it will becomes $v_+ = v_-$ immediately due to the feedback.

v Inverting amplifier

An inverting amplifier consists of an op amp where input voltage v_i is connected to v_- through an input resistance R_i (current from v_i to v_-), v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistance R_f (current from v_- to v_o).

$$v_+ = v_- = 0$$

$$i_i = \frac{v_i}{R_i} = i_f = -\frac{v_o}{R_f}$$

$$v_o = -\frac{R_f}{R_i} v_i$$

The closed-loop gain G is defined as

$$G = \frac{v_o}{v_i} = -\frac{R_f}{R_i}$$

The feedback factor β is defined as

$$\beta = \frac{1}{1 + \frac{R_f}{R_i}}$$

Effect of finite open-loop gain:

$$v_+ = 0$$

$$v_o = -Av_-$$

$$i_i = \frac{v_i - v_-}{R_i} = \frac{v_i + \frac{v_o}{A}}{R_i} = i_f = -\frac{v_o + \frac{v_o}{A}}{R_f}$$

$$G = \frac{v_o}{v_i} = \frac{-\frac{R_f}{R_i}}{1 + \frac{1+\frac{R_f}{R_i}}{A}} = \frac{G_{\text{ideal}}}{1 + \frac{1-G_{\text{ideal}}}{A}}$$

$$\lim_{A\to\infty} G = -\frac{R_f}{R_i} = G_{\text{ideal}}$$

For single-pole model $A(s) = \frac{A_0}{1 + \frac{s}{\omega_b}}$ with $A_0 \gg 1 + \frac{R_f}{R_i}$ and $\omega_t = A_0\omega_b$:

$$\begin{aligned} G &= \frac{-\frac{R_f}{R_i}}{1 + \frac{1+\frac{R_f}{R_i}}{A}} \\ &= \frac{-\frac{R_f}{R_i}}{1 + \frac{1+\frac{R_f}{R_i}}{A_0} + \frac{\frac{s}{\omega_t}}{1+\frac{R_f}{R_i}}} \\ &\approx \frac{-\frac{R_f}{R_i}}{1 + \frac{\frac{s}{\omega_t}}{1+\frac{R_f}{R_i}}}, \end{aligned}$$

which is a low-pass STC filter with 3 dB bandwidth $\frac{\omega_t}{1 + \frac{R_f}{R_i}} = \omega_t\beta$ and DC gain $-\frac{R_f}{R_i}$.

vi Transresistance amplifier

A transresistance amplifier consists of an op amp where input current i_i is connected to v_-, v_+ is connected to ground, and v_o is fed back to v_- through two resistances R_o (current from x to v_o) and R_f (current from v_- to x) in series connection, with the node x between R_o and R_f connected to ground through a resistance R_g (current from x to ground).

$$v_x = -R_f i_i$$

$$i_g = -\frac{R_f}{R_g} i_i$$

$$i_o = \left(1 + \frac{R_f}{R_g}\right) i_i$$

$$v_o = -R_f i_i - R_o i_o = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right) i_i$$

$$\frac{v_o}{i_i} = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right)$$

vii Weighted summing amplifier

A weighted summing amplifier consists of an op amp, input voltages $v_1, v_2, \ldots, v_n$, n input resistors $R_1, R_2, \ldots, R_n$, and a feedback resistor R_f, where for any $k \in \mathbb{N} \wedge k \leq n$, v_k is connected to v_- through R_k, v_+ is connected to ground, and v_o is fed back to v_- through R_f.

$$i_k = \frac{v_k}{R_k}$$

$$\sum_{k=1}^{n} i_k = i_f = \frac{-v_o}{R_f}$$

$$v_o = -R_f \sum_{k=1}^{n} \frac{v_k}{R_k}$$

When $R_k = R$ for all $k \in \mathbb{N} \wedge k \leq n$:

$$v_o = -\frac{R_f}{R} \sum_{k=1}^{n} v_k$$

viii Non-inverting amplifier

A non-inverting (or noninverting) amplifier consists of an op amp where input voltage v_i is connected to v_+, v_- is connected ground through a ground resistance R_g (current from v_- to ground), and v_o is fed back to v_- through a feedback resistance R_f (current from v_o to v_-).

$$v_+ = v_- = v_i$$

$$i_g = \frac{v_i}{R_g} = i_f = \frac{v_o - v_i}{R_f}$$

$$v_o = \left(1 + \frac{R_f}{R_g}\right) v_i$$

The closed-loop gain G is defined as

$$G = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_g}$$

The feedback factor β is defined as

$$\beta = \frac{1}{1 + \frac{R_f}{R_g}}$$

Effect of finite open-loop gain:

$$v_+ = v_i$$

$$v_o = A(v_i - v_-)$$

$$v_- = v_i - \frac{v_o}{A}$$

$$i_g = \frac{v_-}{R_g} = \frac{v_i - \frac{v_o}{A}}{R_g} = i_f = \frac{v_o - v_-}{R_f} = \frac{v_o - v_i + \frac{v_o}{A}}{R_f}$$

$$G = \frac{v_o}{v_i} = \frac{1 + \frac{R_f}{R_g}}{1 + \frac{1 + \frac{R_f}{R_g}}{A}} = \frac{G_{\text{ideal}}}{1 + \frac{G_{\text{ideal}}}{A}}$$

$$\lim_{A \to \infty} G = 1 + \frac{R_f}{R_g} = G_{\text{ideal}}$$

For single-pole model $A(s) = \frac{A_0}{1 + \frac{s}{\omega_b}}$ with $A_0 \gg 1 + \frac{R_f}{R_g}$ and $\omega_t = A_0 \omega_b$:

$$\begin{aligned} G &= \frac{1 + \frac{R_f}{R_g}}{1 + \frac{1 + \frac{R_f}{R_g}}{A}} \\ &= \frac{1 + \frac{R_f}{R_g}}{1 + \frac{1 + \frac{R_f}{R_g}}{A_0} + \frac{\frac{s}{\omega_t}}{1 + \frac{R_f}{R_g}}} \\ &\approx \frac{1 + \frac{R_f}{R_g}}{1 + \frac{\frac{s}{\omega_t}}{1 + \frac{R_f}{R_g}}} \end{aligned}$$

which is a low-pass STC filter with 3 dB bandwidth $\frac{\omega_t}{1 + \frac{R_f}{R_g}} = \omega_t \beta$ and DC gain $1 + \frac{R_f}{R_g}$.

ix Unity gain amplifier, unity gain buffer, buffer amplifier, or voltage follower

A unity gain amplifier consist of an op amp where input voltage v_i is connected to v_+, and v_o is fed back to v_-.

$$v_o = v_- = v_+ = v_i$$

x Difference amplifier or differential amplifier

A difference amplifier consists of an op amp where first input voltage v_1 is connected to v_- through an input resistance R_1 (current from v_1 to v_-), second input voltage v_2 is connected to v_+ through an input resistance R_2 (current from v_2 to v_+), v_+ is connected to ground through a ground resistance R_g (current from v_+ to ground), and v_o is fed back to v_- through a feedback resistance R_f (current from v_- to v_o). Denote common-mode signal $v_{cm} = \frac{v_1 + v_2}{2}$ and differential signal $v_d = v_2 - v_1$.

$$v_o = -\frac{R_f}{R_1} v_1 + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1}\right) v_2$$

$$i_1 = \frac{1}{R_1} \left(v_1 - \frac{R_g}{R_2 + R_g} v_2\right)$$

$$i_2 = \frac{v_2}{R_2 + R_g}$$

Common-mode gain A_{cm} and differential gain A_d, defined by

$$v_o = v_{cm} A_{cm} + v_d A_d,$$

are

$$A_{cm} = \frac{R_g}{R_2 + R_g}\left(1 - \frac{R_2}{R_g}\frac{R_f}{R_1}\right)$$

$$A_d = \frac{1}{2}\frac{R_g}{R_2 + R_g}\left(1 + \frac{R_2 + 2R_g}{R_g}\frac{R_f}{R_1}\right)$$

Common-mode rejection ratio (CMRR) is defined as

$$\text{CMRR} = 20 \log \left|\frac{A_d}{A_{cm}}\right|$$

To make $A_{cm} = 0$, we need

$$\frac{R_f}{R_1} = \frac{R_g}{R_2},$$

at which

$$A_d = \frac{R_f}{R_1} = \frac{R_g}{R_2}$$

Differential input resistance R_d is

$$R_d = R_1 + R_2$$

Thus, high differential gain hurts differential input resistance, which lower the voltage applied to the amplifier since source has internal resistance.

Proof. Use superposition. With v_1 enabled and v_2 disabled,

$$v_+ = v_- = 0$$

$$i_2 = i_g = 0$$

$$i_1 = \frac{v_1}{R_1} = i_f$$

$$v_o = -i_f R_f = -\frac{R_f}{R_1} v_1$$

With v_2 enabled and v_1 disabled,

$$i_2 = i_g = \frac{v_2}{R_2 + R_g}$$

$$v_+ = v_- = i_g R_g = \frac{R_g}{R_2 + R_g} v_2$$

$$i_1 = -\frac{v_-}{R_1} = -\frac{R_g}{R_1(R_2 + R_g)} v_2 = i_f$$

$$v_o = v_- - i_f R_f = \frac{R_g}{R_2 + R_g} v_2 + \frac{R_g R_f}{R_1(R_2 + R_g)} v_2 = \frac{R_g}{R_2 + R_g}\left(1 + \frac{R_f}{R_1}\right) v_2$$

Sum:

$$v_o = -\frac{R_f}{R_1} v_1 + \frac{R_g}{R_2 + R_g}\left(1 + \frac{R_f}{R_1}\right) v_2$$

$$i_1 = \frac{v_1}{R_1} - \frac{R_g}{R_1(R_2+R_g)}v_2 = \frac{1}{R_1}\left(v_1 - \frac{R_g}{R_2+R_g}v_2\right)$$

$$i_2 = \frac{v_2}{R_2+R_g}$$

$$v_1 = v_{cm} - \frac{v_d}{2}$$

$$v_2 = v_{cm} + \frac{v_d}{2}$$

$$\begin{aligned} v_o &= -\frac{R_f}{R_1}\left(v_{cm} - \frac{v_d}{2}\right) + \frac{R_g}{R_2+R_g}\left(1+\frac{R_f}{R_1}\right)\left(v_{cm}+\frac{v_d}{2}\right) \\ &= \frac{R_g}{R_2+R_g}\left(1-\left(\frac{R_2+R_g}{R_g}-1\right)\frac{R_f}{R_1}\right)v_{cm} + \frac{R_g}{R_2+R_g}\left(1+\left(\frac{R_2+R_g}{R_g}+1\right)\frac{R_f}{R_1}\right)\frac{v_d}{2} \\ &= \frac{R_g}{R_2+R_g}\left(1-\frac{R_2}{R_g}\frac{R_f}{R_1}\right)v_{cm} + \frac{R_g}{R_2+R_g}\left(1+\frac{R_2+2R_g}{R_g}\frac{R_f}{R_1}\right)\frac{v_d}{2} \end{aligned}$$

$$A_{cm} = \frac{R_g}{R_2+R_g}\left(1-\frac{R_2}{R_g}\frac{R_f}{R_1}\right)$$

$$A_d = \frac{1}{2}\frac{R_g}{R_2+R_g}\left(1+\frac{R_2+2R_g}{R_g}\frac{R_f}{R_1}\right)$$

To make $A_{cm} = 0$, we need

$$1 = \frac{R_2}{R_g}\frac{R_f}{R_1}$$

$$\frac{R_f}{R_1} = \frac{R_g}{R_2}$$

at which

$$\begin{aligned} A_d &= \frac{1}{2}\frac{R_g}{R_2+R_g}\left(1+\frac{R_2+2R_g}{R_g}\frac{R_f}{R_1}\right) \\ &= \frac{1}{2}\frac{R_g}{R_2+R_g}\left(1+\frac{R_2+2R_g}{R_2}\right) \\ &= \frac{R_g}{R_2+R_g}\frac{R_2+R_g}{R_2} \\ &= \frac{R_g}{R_2} \end{aligned}$$

Differential input resistance:

$$i_d = \frac{v_+}{R_g} = \frac{v_2 - v_+}{R_2} = \frac{v_+ - v_2 + v_d}{R_1}$$

$$R_2 v_+ = R_g v_2 - R_g v_+$$

$$R_1 v_2 - R_1 v_+ = R_2 v_+ - R_2 v_2 + R_2 v_d$$

$$v_2 = \left(\frac{R_2+R_g}{R_g}\right)v_+$$

$$(R_1+R_2)(v_2 - v_+) = R_2 v_d$$

$$v_+ = \frac{R_g}{R_1+R_2}v_d$$

$$i_d = \frac{1}{R_1 + R_2} v_d$$
$$R_d = \frac{v_d}{i_d} = R_1 + R_2$$

□

xi Instrumentation amplifier (INA)

An instrumentation amplifier consists of a difference amplifier (second stage) with outputs of two identical non-inverting amplifiers (first stage) as inputs, which behaves the same as the difference amplifier but with both common-mode and differential gains multiplied by the gain of the first stage. It can achieve higher differential gain than difference amplifier under same differential input resistance. However, common-mode signal is also amplified in the first stage, so CMRR of second stage must be good, i.e., the resistances must be precise.

xii Inverting configuration with general impedance

Consider an op amp where input voltage v_i is connected to v_- through an input impedance Z_i (current from v_i to v_-), v_+ is connected to ground, and v_o is fed back to v_- through a feedback impedance Z_f (current from v_- to v_o).

Laplace domain:

$$V_+ = V_- = 0$$
$$I_i = \frac{V_i}{Z_i} = I_f = -\frac{V_o}{Z_f}$$
$$V_o = -\frac{Z_f}{Z_i} V_i$$

xiii Inverting integrator or Miller integrator

An inverting integrator consists of an op amp where input voltage v_I is connected to v_- through an input resistance R (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a feedback capacitance C (current from v_- to v_O).

$v_O(0^-)$ is the initial voltage across C (v_O relative to v_-).

Time domain:

$$v_+ = v_- = 0$$
$$i_I = \frac{v_I}{R} = i_C = -C\frac{\mathrm{d}v_O}{\mathrm{d}t}$$
$$v_O = v_O(0^-) - \frac{1}{RC}\int_0^t v_I(\tau)\,\mathrm{d}\tau$$

Laplace domain:

$$V_+ = V_- = 0$$
$$I_I = \frac{V_I}{R} = I_C = -CsV_O + Cv_O(0^-)$$
$$V_O = \frac{1}{s} v_O(0^-) - \frac{1}{RCs} V_I$$

Transfer function: Infinite DC gain. Thus, any tiny DC component in the input signal will blow this out. $\frac{\pi}{2}$ phase shift.

$$H(s) = \frac{V_o}{V_i} = -\frac{1}{RCs}$$

Unity-gain angular frequency:

$$\frac{1}{RC}$$

xiv Inverting integrator or Miller integrator feasible configuration

An inverting integrator feasible configuration consists of an op amp where input voltage v_I is connected to v_- through an input resistance R (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a feedback capacitance C and feedback resistance R_f in parallel connection with the sum of their current be i_f (from v_- to v_O).

$v_O(0^-)$ is the initial voltage across C (v_O relative to v_-).

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_f = -\left(\frac{1}{R_f} + Cs\right)V_O + Cv_O(0^-) = -\frac{1 + R_f Cs}{R_f}V_O + Cv_O(0^-)$$

$$V_O = \frac{R_f C}{1 + R_f Cs}v_O(0^-) - \frac{\frac{R_f}{R}}{1 + R_f Cs}V_I$$

Transfer function: STC low-pass with cutoff frequency $\frac{1}{R_f C}$ and DC gain $\frac{R_f}{R}$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R_f}{R}}{1 + R_f Cs}$$

xv Inverting differentiator

An inverting differentiator consists of an op amp where input voltage v_I is connected to v_- through an input capacitor C (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a feedback resistor R (current from v_- to v_O).

$v_I(0^-)$ is the initial voltage across C (v_I relative to v_-).

Time domain:

$$i_I = C\frac{\mathrm{d}v_I}{\mathrm{d}t} = i_f = \frac{-v_O}{R}$$

$$v_O = -RC\frac{\mathrm{d}v_I}{\mathrm{d}t}$$

Laplace domain:

$$I_I = CsV_I - Cv_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -RCsV_I + RCv_I(0^-)$$

Transfer function: Infinite gain at $\omega = \infty$. $-\frac{\pi}{2}$ phase shift.

$$H(s) = \frac{V_o}{V_i} = -RCs$$

Unity-gain angular frequency:

$$\frac{1}{RC}$$

xvi Inverting differentiator feasible configuration

An inverting differentiator feasible configuration consists of an op amp where input voltage v_I is connected to v_- through an input capacitor C and a input resistor R_I in series connection (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a feedback resistor R (current from v_- to v_O).

$v_I(0^-)$ is the initial voltage across C (v_I relative to v_-).

Laplace domain:

$$I_I = \frac{1}{R_I + \frac{1}{Cs}} V_I - Cv_I(0^-) = \frac{\frac{s}{R_I}}{s + \frac{1}{R_I C}} V_I - Cv_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -\frac{\frac{R}{R_I}s}{s + \frac{1}{R_I C}} V_I + RCv_I(0^-)$$

Transfer function: STC high-pass with cutoff frequency $\frac{1}{R_I C}$ and gain $\frac{R}{R_I}$ at $\omega = \infty$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R}{R_I}s}{s + \frac{1}{R_I C}}$$

xvii Large-signal operation

- Output voltage saturation: A real op amp operates linearly over a limited range of output voltages and saturates if the output voltage reaches a saturation voltage, which is related to the positive and negative power supply voltages. An op amp with saturation voltage equal to its power supplies is said to offer rail-to-rail output.
- Output current limits: A real op amp's output current is limited to a specified maximum. This also imposes a output voltage limit equal to the maximum output current multiplied by the sum of output impedance and load impedance. For 741, its maximum output current is 20 mA.

xviii Slew rate (SR)

There is a specific maximum rate of change possible at the output voltage of a real op amp, called slew rate (SR),

$$\mathrm{SR} = \left| \frac{\mathrm{d}v_o}{\mathrm{d}t} \right| \Bigg|_{\max},$$

usually specified on the op-amp data sheet in units of V/μs.

To measure it, connect v_o to v_- and v_+ to a voltage source v_t and measure the rate of change of v_o before it reaches v_t. The op amp is then said to be slewing, and the output is slew-rete limited.